Non-linear Time series analysis - CE3

Rujia Li, Xinxu Yao

November 2025

1 Part 1: Simulation and discretization of diffusion processes

Question 1a: Simulation of the Bonhoeffer-Van der Pol equations

In this section, we investigate the stochastic behavior of the Bonhoeffer-Van der Pol (BVP) system. The system describes the firing of a single neuron, incorporating additive noise to simulate membrane imperfections. The continuous system is discretized using the Euler-Maruyama approximation.

The two-dimensional stochastic difference equations used for the simulation are given by:

$$Y_{n+1}^1 = Y_n^1 + \theta_3 \left(Y_n^1 + Y_n^2 - \frac{1}{3}(Y_n^1)^3 + \theta_4 \right) \Delta + \sigma\sqrt{\Delta} Z_{n+1} \tag{1}$$

$$Y_{n+1}^2 = Y_n^2 - \frac{1}{\theta_3} \left(Y_n^1 + \theta_2 Y_n^2 - \theta_1 \right) \Delta \tag{2}$$

where $Z_{n+1} \sim N(0,1)$ represents the standard normal random variable derived from the Wiener increment ΔW_{n+1}. The simulation parameters are set as follows: $\Delta = 2^{-9}$, $T = 100$, $\theta_1 = 0.7$, $\theta_2 = 0.8$, $\theta_3 = 3.0$, and $\theta_4 = -0.34$. The initial conditions are initialized at $(Y_0^1, Y_0^2) = (-1.9, 1.2)$.

We performed simulations for five different noise levels: $\sigma \in \{0, 0.10, 0.20, 0.30, 0.40\}$. The time series of the state variables Y^1 and Y^2, along with their corresponding phase plots, are presented below.

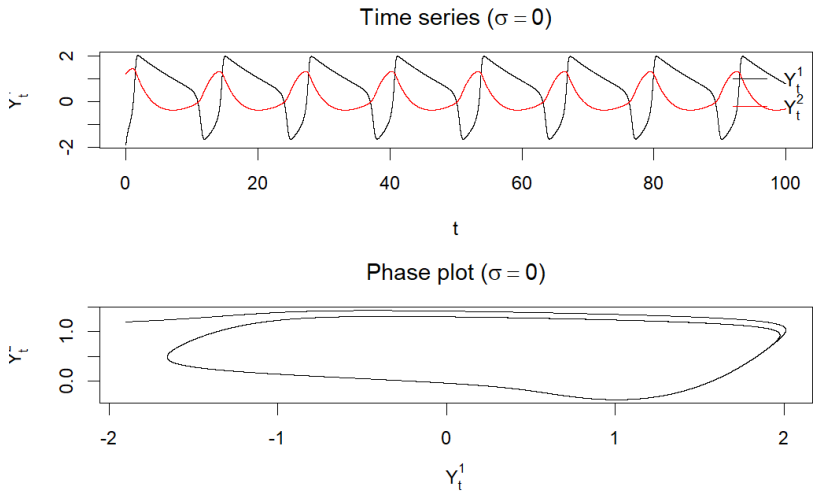

Figure 1: Simulation results for $\sigma = 0$ (Deterministic case). Top: Time series. Bottom: Phase plot showing a perfect limit cycle.

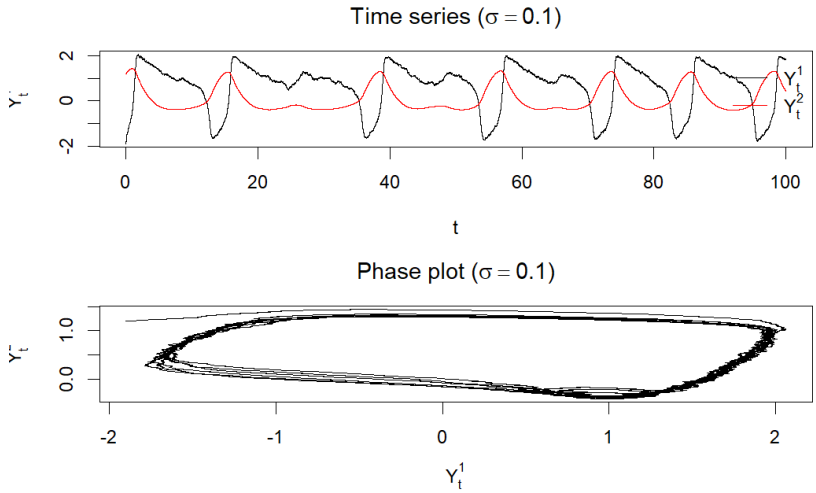

Figure 2: Simulation results for $\sigma = 0.10$. The trajectory shows slight stochastic perturbations around the limit cycle.

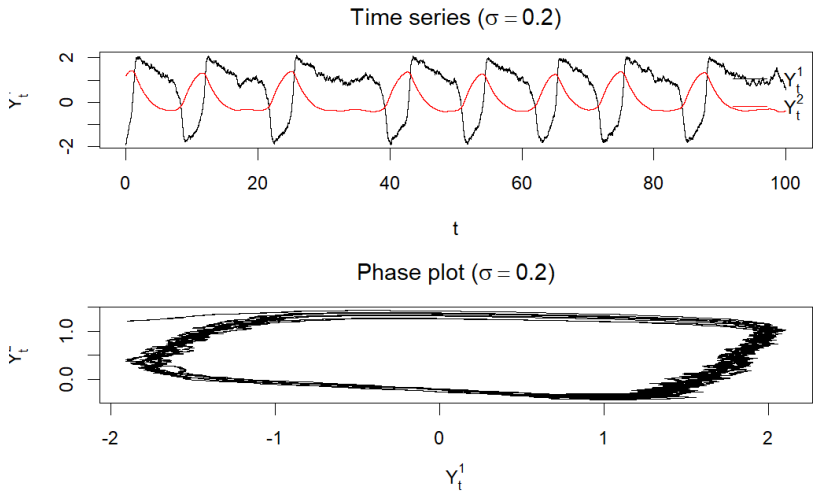

Figure 3: Simulation results for $\sigma = 0.20$. Noise becomes more visible in the phase plot "tube".

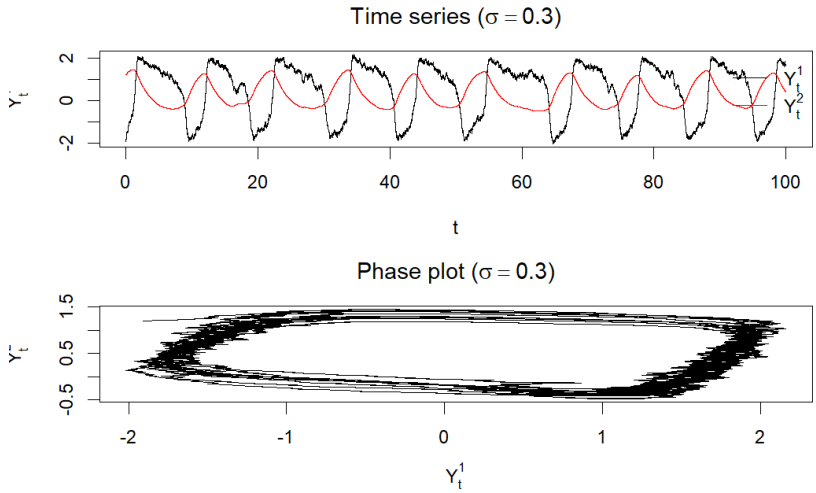

Figure 4: Simulation results for $\sigma = 0.30$. Significant deviations from the deterministic path are observed.

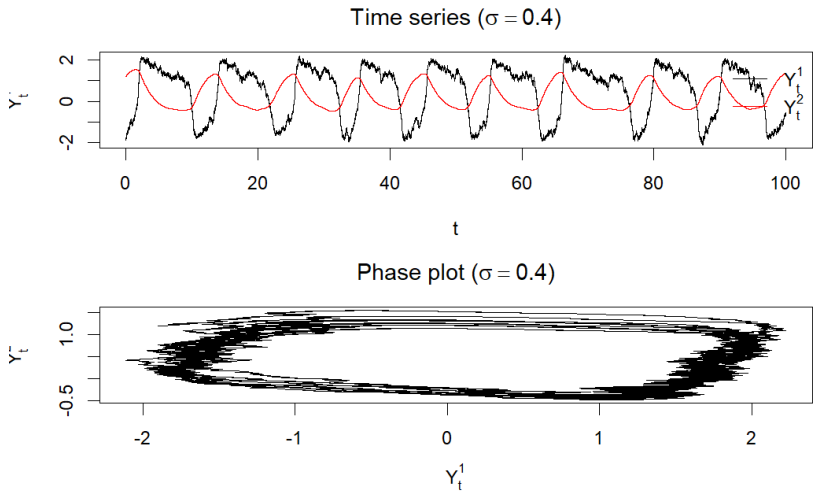

Figure 5: Simulation results for $\sigma = 0.40$. High noise level resulting in a broad stochastic band in the phase space.

Analysis of Noise Effects

Based on the visual inspection of the time series and phase plots across the different values of σ, we observe the following behaviors:

- **Deterministic Case ($\sigma = 0$):** As shown in Figure 1, the system exhibits a stable limit cycle. The phase plot is a closed, smooth loop, indicating a stable periodic orbit.
- **Low Noise ($\sigma = 0.10 - 0.20$):** Introducing small additive noise (Figures 2 and 3) perturbs the trajectory.In time series plots, the Yt becomes unstable during the period. In the phase space, the strictly 1-dimensional curve of the limit cycle broadens into a narrow "tube." The system still stays very close to the deterministic limit cycle.
- **High Noise ($\sigma = 0.30 - 0.40$):** As the noise intensity increases (Figures 4 and 5), the stochastic term $\sigma \Delta W$ dominates more of the local dynamics. The phase plot widens significantly, representing a probability distribution of trajectories rather than a single path. In the time series, the amplitude of the spikes and the inter-spike intervals fluctuate more noticeably. Despite the high noise, the system does not diverge or collapse to a fixed point; it continues to oscillate. This shows that the system's natural tendency to cycle (the deterministic attractor) is powerful enough to keep the trajectory.

In conclusion, the Bonhoeffer-Van der Pol system is robust. While the noise introduces variability in the specific path and timing of the neural firing (phase

diffusion), the fundamental oscillatory behavior—representing the charging and discharging of the neuron—persists even at higher noise levels ($\sigma = 0.40$).

Question 1b: Empirical Density of the Stationary Solution

In this section, we move from analyzing individual trajectories to approximating the stationary distribution of the stochastic system. Although the Euler-Maruyama method is a weak approximation, it allows us to visualize the asymptotic behavior of the system's moments and density.

We simulated the system for $T = 100$ (reusing the parameters from Question 1a) for $\sigma \in \{0.10, 0.20, 0.30, 0.40\}$. The phase plane was partitioned into a 100×100 grid. By counting the number of time steps the trajectory spends in each cell, we constructed 3D histograms that approximate the stationary probability density function (PDF).

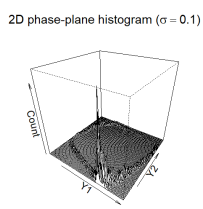

Figure 6: *
$\sigma = 0.10$

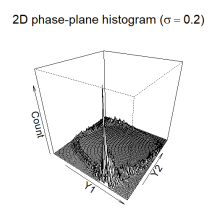

Figure 7: *
$\sigma = 0.20$

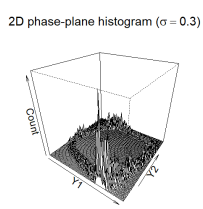

Figure 8: *
$\sigma = 0.30$

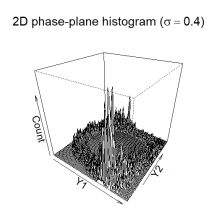

Figure 9: *
$\sigma = 0.40$

Figure 10: 3D histograms of the phase plane trajectories. The vertical axis (Count) represents the unnormalized empirical probability density.

Comparison with Standard Phase Plots

The standard phase plots in Question 1a strictly visualize the *support* of the trajectory—showing geometrically where the system can exist in the (Y^1, Y^2) plane. However, the 3D histograms in Figure 10 provide significant additional information:

1. **Probability Density and Time Allocation:** The "height" of the histogram (z-axis) corresponds to the amount of time the system spends in a specific region. This effectively approximates the invariant measure of the system. A higher peak indicates that the system evolves slowly through that region, while a lower ridge indicates fast dynamics where the system transitions rapidly.

2. **Visualizing Stochastic Spread:**

   - For **low noise** ($\sigma = 0.10$), the histogram resembles a sharp, crater-like wall. The density is highly concentrated around the deterministic limit cycle, indicating that the probability of finding the system far from the orbit is negligible.
   - For **high noise** ($\sigma = 0.40$), the "walls" become significantly wider and lower. The sharp peaks smooth out into broader hills. This visualizes the diffusion effect: the stochastic term pushes the state variable away from the deterministic attractor more frequently, flattening the probability distribution.

In the context of the Bonhoeffer-Van der Pol neuron model, the non-uniform height of the ridges is physically meaningful. The higher regions likely correspond to the "slow" charging phase of the neuron, while the lower regions of the loop correspond to the "fast" discharge (spike) phase, where the state variables change too quickly to accumulate a high count in the histogram bins.

2 Part 2: Modelling a room using CTSM-R

2a: 2-state model of Room1 with time-varying solar gain

Initial Analysis: The Problem with Constant A_w

We began by examining the baseline 2-state model for Room 1 (provided in `ExampleRoom1.R`). This model assumes that the effective window area (A_w) is a constant parameter.

However, physically, the position of the sun changes relative to the building throughout the day. This means the angle of incidence and the amount of solar radiation actually entering the room change over time. A constant parameter cannot capture this dynamic behavior. This limitation is clearly visible when we analyze the residuals of the constant model.

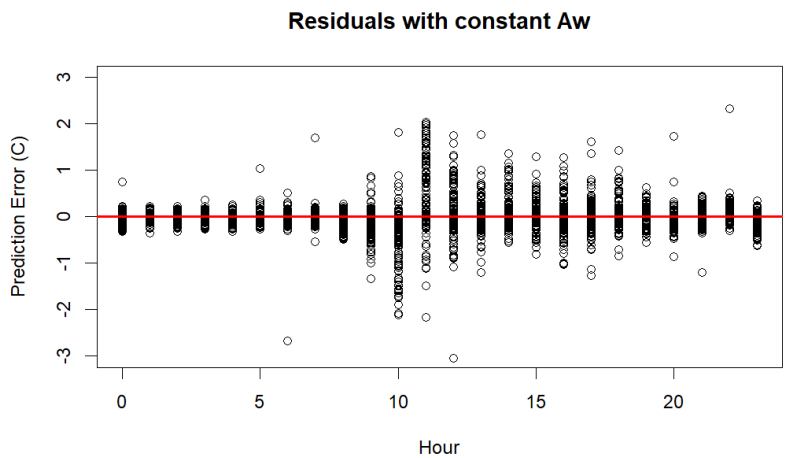

Figure 11: Residuals of the 1-step prediction for the constant A_w model.

As shown in Figure 11, the residuals show high variance and bias around 10:00 AM. This suggests the model fails to capture the rapid temperature changes occurring when the sun hits the east-facing facade in the morning.

Modeling Time-Varying A_w with B-Splines

To address this, we modified the system equation for the indoor temperature (T_i). Instead of a constant A_w, we modeled the effective window area as a time-varying function using B-splines ($bs_k(t)$). We limited the estimation to hours where the sun is active (08:00 to 23:00).

The modified stochastic differential equation for the indoor temperature is:

$$dT_i = \frac{1}{C_i}\left(\frac{1}{R_{ia}}\left(T_a - T_i\right) + \frac{1}{R_{im}}\left(T_m - T_i\right) + \Phi + \left(\sum_{k=1}^{5} a_k bs_k(t)\right) G_y\right) dt + \sigma_1 dw_1 \tag{3}$$

We implemented this in `ctsm` using 5 degrees of freedom for the splines ($df = 5$) to allow flexibility in the shape of the solar gain curve.

Results and Physical Interpretation

After optimizing the parameters, we plotted the estimated effective window area over the course of the day.

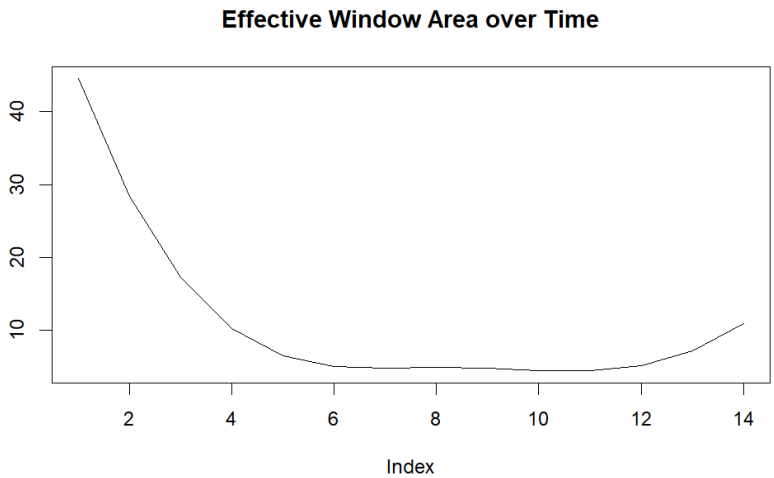

Figure 12: Estimated Effective Window Area (A_w) over time.

The shape of the curve in Figure 12 raises a major concern regarding physical realism. The estimated effective window area reaches a peak of over **40 m^2** in the early morning. Given the physical dimensions of the room, a 40 m^2 window area is impossible.

This indicates that the model is "force-fitting" the data. The optimizer detects a rapid rise in indoor temperature in the morning and, lacking other explanatory variables, attributes all of that heating to solar gain. Consequently, it drastically overestimates the window size.

To further investigate the model fit, we examined the estimated parameters and their statistical significance, shown in Table 1.

Table 1: Parameter estimates for the 2-state model (Task 2a). Initial states (Ti_0, Tm_0) are omitted for brevity.

Parameter	**Estimate**	**Std. Error**	**t value**	**Pr(>|t|)**
Aw_1	44.62	2.710	16.5	$< 2e-16$
Aw_2	$1.48e-7$	$3.04e-5$	0.005	0.996
Aw_3	9.09	1.278	7.11	$1.44e-12$
Aw_4	1.43	1.361	1.05	0.295
Aw_5	10.88	3.818	2.85	0.004
C_i	9.07	0.213	42.5	$< 2e-16$
C_m	$8.36e4$	$4.24e5$	0.197	0.844
R_{ia}	18.76	12.26	1.53	0.126
R_{im}	0.44	0.017	25.3	$< 2e-16$
e_{11}	-21.19	200.5	-0.106	0.916
p_{11}	-1.63	0.020	-80.8	$< 2e-16$
p_{22}	-0.98	0.052	-18.7	$< 2e-16$

Several key observations can be made from the parameter estimates:

- **Statistical Evidence of Overestimation:** The coefficient Aw_1 corresponds to the morning peak and is estimated at 44.62. Its extremely low p-value ($< 2e-16$) confirms that this unrealistic value is statistically "necessary" for this specific model structure to fit the data, reinforcing the conclusion that the model compensates for missing dynamics by inflating solar gain.

- **Physical Consistency:** The heat capacity of the thermal mass ($C_m \approx 8.36 \times 10^4$) is orders of magnitude larger than that of the air ($C_i \approx 9.07$). This relationship is physically consistent, as the building structure stores significantly more energy than the air volume.

- **Insignificant Parameters:** The coefficients Aw_2 ($p = 0.996$) and Aw_4 ($p = 0.295$) are not statistically significant. This suggests that the chosen B-spline basis functions for these time periods do not contribute meaningfully to the model, or that the solar gain is negligible during these specific intervals. Furthermore, the thermal resistance to the ambient (R_{ia}) shows a high p-value (0.126), indicating that the model struggles to accurately identify the heat loss to the outside environment.

Residual Analysis of the New Model

Finally, we inspected the residuals for this new time-varying model to see if the fit improved.

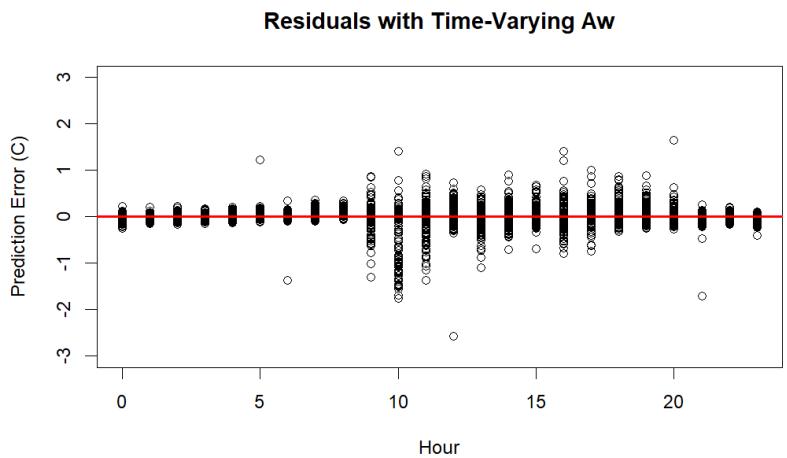

Figure 13: Residuals of the model with time-varying A_w. Significant instability remains at 10:00 AM.

As seen in Figure 13, although we introduced the time-varying parameter, the residuals around 10:00 AM are still quite unstable. The persistence of these errors, combined with the unrealistic estimation of the window area, indicates that simply making A_w time-varying is not enough. The model is likely missing other significant dynamics, which motivates the need for further model improvements in the next task.

2b: Improving the single-room model

In this task, we aimed to address the limitations identified in Task 2a, specifically the unrealistic effective window area ($A_w > 40m^2$) and the residual instability around 10:00 AM. We adopted an iterative approach, testing structural modifications before identifying the root cause of the model mismatch.

Attempt 1: Increasing Spline Flexibility

Our first hypothesis was that the default B-spline basis ($df = 5$) was too stiff to capture the rapid change in solar angle or shading effects in the morning. To address this, we increased the degrees of freedom for the solar splines to 8.

$$dT_i = \cdots + \left(\sum_{k=1}^{8} a_k bs_k(t) \right) G_y dt + \ldots \tag{4}$$

However, this modification did not yield a stable solution.

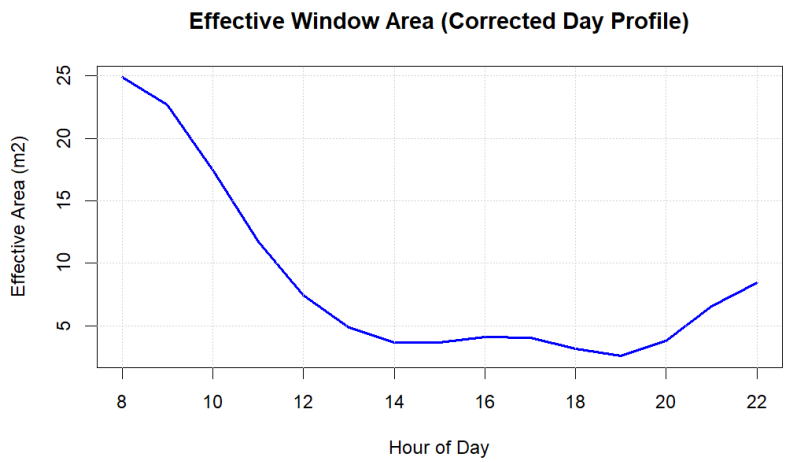

Figure 14: Effective Window Area.

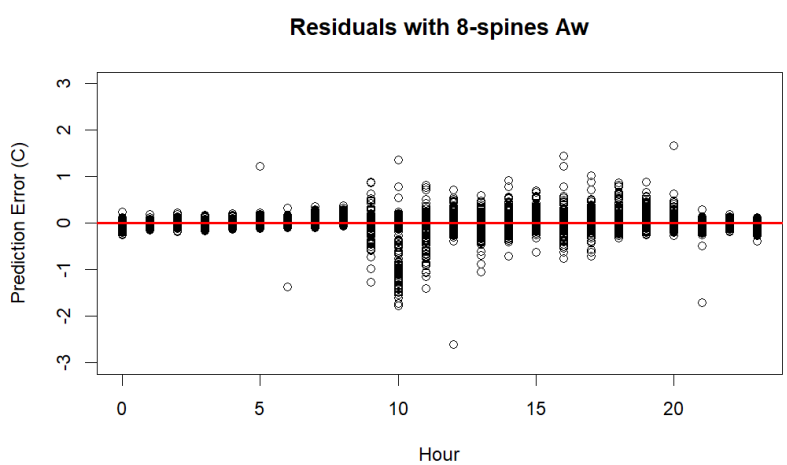

Figure 15: Residuals for the 8-spines model.

- **Overfitting:** The coefficients for the early morning (Aw_1, Aw_2) and late afternoon often hit the lower bounds (0) or behaved erratically, suggesting the model was trying to fit noise rather than signal.
- **Multicollinearity:** Increasing the number of splines increased the correlation between adjacent basis functions, making the parameter estimates highly uncertain.

- **Residuals:** Most importantly, the large residual spikes around 10:00 AM persisted.

We concluded that simply adding mathematical flexibility (more splines) was insufficient and reverted to $df = 5$ to focus on physical improvements.

Attempt 2: Solar Heat Distribution to Thermal Mass

Next, we hypothesized that solar radiation does not exclusively heat the air but also enters the thermal mass (walls/floor) directly. We introduced a distribution parameter $p_{solar} \in [0, 1]$ representing the fraction of solar energy entering the air directly.

The modified system equations are:

$$dT_i = \frac{1}{C_i}\left(\frac{1}{R_{ia}}(T_a - T_i) + \frac{1}{R_{im}}(T_m - T_i) + \Phi + p_{solar} \cdot A_w(t)G_y\right)dt + \sigma_1 dw_1 \tag{5}$$

$$dT_m = \frac{1}{C_m}\left(\frac{1}{R_{im}}(T_i - T_m) + (1 - p_{solar}) \cdot A_w(t)G_y\right)dt + \sigma_2 dw_2 \tag{6}$$

We conducted a series of experiments to test this hypothesis, but the results consistently contradicted physical laws:

- **Unconstrained Estimation:** Initially, we allowed p_{solar} to be estimated freely. The model converged to $p_{solar} > 0.99$, implying that nearly all solar energy heats the air directly. This contradicts the physical reality that sunlight hits surfaces (walls/floors) first.

- **Constraining Solar Distribution (p_{solar}):** To enforce physical realism, we fixed p_{solar} to a smaller value (forcing heat into the thermal mass). However, the optimizer responded by reducing the heat capacity of the thermal mass (C_m) to an extremely low value (≈ 60). This implied the concrete walls had the heat storage capacity of a sheet of paper, effectively removing the "thermal mass" effect to explain the rapid temperature rise.

- **Constraining Heat Capacity (C_m):** Finally, we fixed C_m to a physically reasonable value to prevent the previous issue. Under these conditions, the model hit the upper bound for the window area (Aw_1 reached the limit of $25m^2$), and the large residual spike at 10:00 AM persisted.

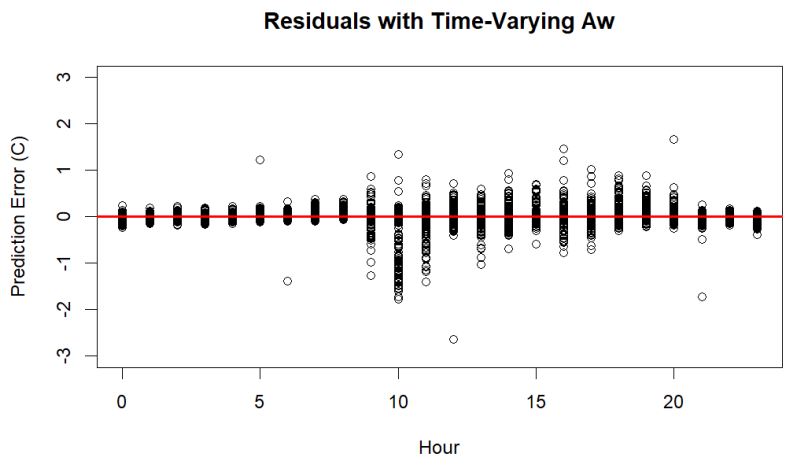

Figure 16: Residuals for the Solar Distribution model (Attempt 2). Even with complex adjustments, the fit remains poor around 10:00 AM.

Conclusion: The model desperately attempted to attribute the morning temperature rise to solar radiation—either by forcing all heat into the air, deleting the thermal mass, or inflating the window area. Since none of these are physically permissible, we concluded that "solar gain" is *not* the cause of the morning temperature spike. We must look for an alternative heat source which is not included in this model.

Hypothesis Verification: The "Student Effect" and Weekend Calibration

To investigate the root cause of the model mismatch observed in previous attempts, we visualized the measured indoor temperature (yTi) over the entire data collection period.

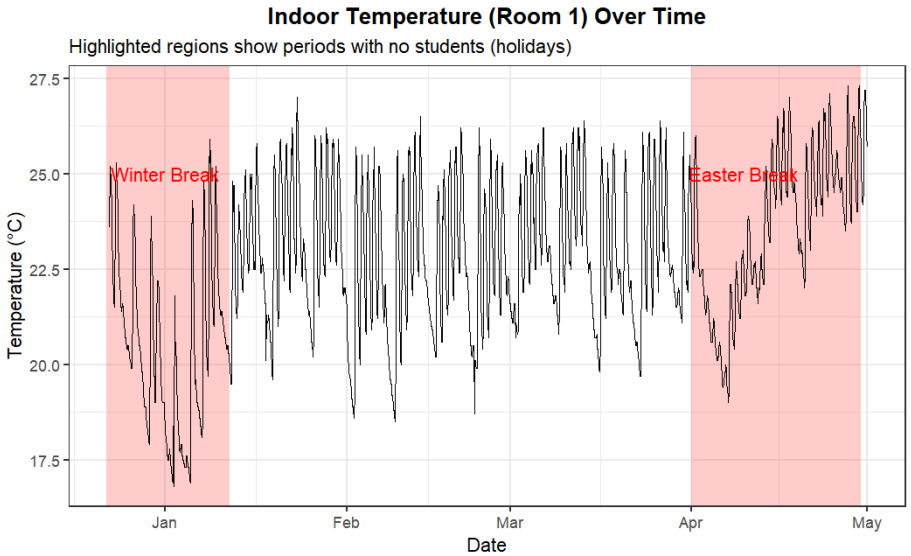

Figure 17: Measured indoor temperature over the entire dataset. Distinct dynamic patterns are visible during holiday periods compared to the semester.

As illustrated in Figure 17, the temperature dynamics exhibit a clear shift in behavior during specific intervals, most notably from December 22 to January 12 and from April 1 to April 30. During these periods (which correspond to Winter and Easter breaks), the temperature evolution appears different from the rest of the time.

Based on this observation, we hypothesize that the unmodeled heat source causing the morning instability is **stochastic human activity** (the "Student Effect"). When students are present, they introduce significant uncertainty into the system through:

- **Internal Heat Gains:** Increased heat load from body heat and the use of electrical appliances (laptops, laboratory equipment).
- **Ventilation Randomness:** Unpredictable behavior such as opening and closing windows, which drastically changes the air exchange rate and thermal dynamics.

To verify this hypothesis and to identify the true physical properties of the building without this human interference, we adopted a **two-stage modeling strategy**.

Step 1: Calibration on Weekend Data

We first fitted a model using **only weekend data**. Since we suspected the radiator heat transfer might also be complex, we improved the model structure by introducing a distribution parameter $p_{ph} \in [0, 1]$. This allows the radiator power Φ to heat both the air (T_i) and the thermal mass (T_m).

The calibrated "Weekend Model" is given by:

$$dT_i = \frac{1}{C_i}\left(\frac{1}{R_{ia}}(T_a - T_i) + \frac{1}{R_{im}}(T_m - T_i) + p_{ph}\Phi + A_w(t)G_y\right)dt + \sigma_1 dw_1 \tag{7}$$

$$dT_m = \frac{1}{C_m}\left(\frac{1}{R_{im}}(T_i - T_m) + (1 - p_{ph})\Phi\right)dt + \sigma_2 dw_2 \tag{8}$$

Table 2: Parameter estimates for the Weekend Calibration Model. These values are used as fixed priors for the final model.

Parameter	**Estimate**	**Std. Error**	**t value**	**Pr($>\|t\|$)**
Aw_1	3.53	3.75	0.94	0.347
Aw_2	4.96	1.57	3.15	0.002
Aw_3	$1.11e^{-12}$	$9.17e^{-11}$	0.01	0.990
Aw_4	9.38	3.27	2.87	0.004
Aw_5	$7.35e^{-12}$	$6.08e^{-10}$	0.01	0.990
C_i	12.74	1.47	8.66	$< 2e^{-16}$
C_m	$1.63e^5$	$8.25e^4$	1.98	0.048
R_{ia}	19.75	1.82	10.84	$< 2e^{-16}$
R_{im}	0.46	0.076	6.13	$1.86e^{-9}$
p_{ph}	0.99	0.065	15.20	$< 2e^{-16}$
e_{11}	-5.63	0.10	-55.95	$< 2e^{-16}$
p_{11}	-9.74	0.26	-37.22	$< 2e^{-16}$
p_{22}	-1.38	0.11	-12.54	$< 2e^{-16}$

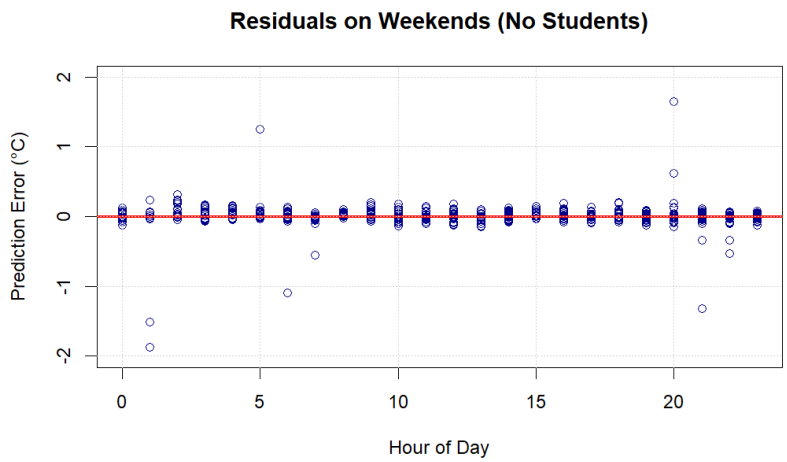

Figure 18: Model residuals when fitted to weekend data only. The variance is stable, and the physical parameters (C, R, p_{ph}) converged to realistic values.

The results (Figure 18) confirmed our hypothesis. The residuals on weekends were well-behaved, and the estimated parameters (e.g., C_m, A_w) represented the building's true physics. We treated these estimates as the ground truth.

The Final Model: Fixed Priors and Occupancy Estimation

Step 2: Full Model with Fixed Physics

For the final model covering the entire dataset, we applied a "Fixed Prior" strategy. We assumed that the building's envelope (C, R) and solar geometry (A_w) do not change during the week.

Therefore, we "locked" the physical parameters $(C_i, C_m, R_{ia}, R_{im}, p_{ph})$ and the solar spline coefficients (A_w) to the values estimated in Step 1. These parameters were constrained within extremely narrow bounds in the optimizer.

The only free variable introduced was the "Occupancy Heat" (Q_{occ}), modeled by 5 B-splines active during school hours (09:00–16:00). The final system structure is:

$$dT_i = \frac{1}{C_i}\left(\ldots \text{Fixed Physics} \cdots + Q_{occ}(t)\right) dt + \sigma_1 dw_1 \tag{9}$$

$$dT_m = \frac{1}{C_m}\left(\ldots \text{Fixed Physics} \ldots\right) dt + \sigma_2 dw_2 \tag{10}$$

where Fixed Physics refers to the terms identified in Step 1.

Table 3: Parameter estimates for the initial full model, Note the extremely low value of e_{11}, suggesting overfitting.

Parameter	**Estimate**	**Std. Error**	**t value**	**Pr(>|t|)**
Ti_0	21.70	0.16	139.74	$< 2e^{-16}$
Tm_0	20.90	0.66	31.45	$< 2e^{-16}$
Aw_1	3.53	0.007	479.02	$< 2e^{-16}$
Aw_2	4.96	0.010	486.35	$< 2e^{-16}$
Aw_3	-0.009	0.025	-0.34	0.730
Aw_4	9.38	0.008	1205.41	$< 2e^{-16}$
Aw_5	0.004	0.017	0.24	0.808
C_i	12.74	0.012	1039.68	$< 2e^{-16}$
C_m	$1.63e^5$	3.67	44502.6	$< 2e^{-16}$
R_{ia}	19.75	0.008	2417.43	$< 2e^{-16}$
R_{im}	0.46	0.001	381.46	$< 2e^{-16}$
p_{ph}	0.998	0.006	173.38	$< 2e^{-16}$
e_{11}	$\mathbf{-16.70}$	45.31	-0.37	0.713
p_{11}	-1.52	0.027	-55.41	$< 2e^{-16}$
p_{22}	-1.19	0.134	-8.91	$< 2e^{-16}$
Q_1	9.97	0.080	124.89	$< 2e^{-16}$
Q_2	6.79	0.773	8.78	$< 2e^{-16}$
Q_3	3.94	0.994	3.96	$7.75e^{-5}$
Q_4	0.85	0.725	1.17	0.242
Q_5	3.04	0.450	6.75	$2.06e^{-11}$

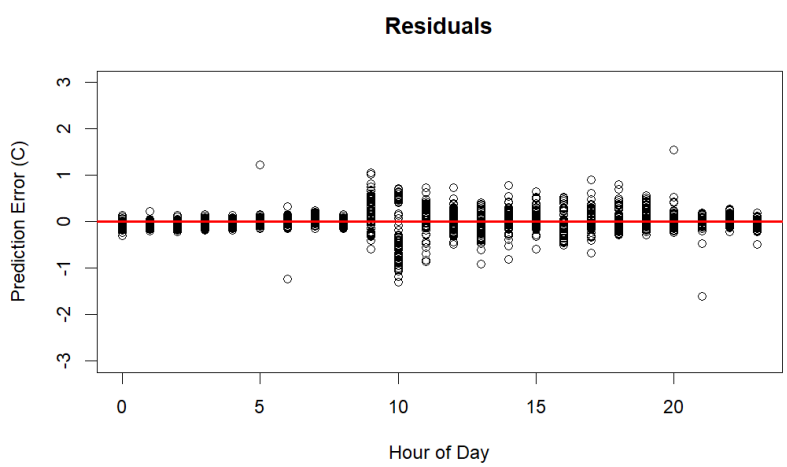

Figure 19: Residuals of the final model with Occupancy term (Q_{occ}). The systematic bias is significantly reduced compared to Task 2a.

As observed in Figure 19, some residual instability around 10:00 AM persists. This can be explained by the inherent stochasticity of student activities—such as random window opening or varying occupancy levels—which introduces irregular noise that smooth spline functions cannot fully capture. However, compared to Task 2a, the systematic bias has been eliminated, and the model results have been significantly improved.

We noticed the initial estimate for e_{11} was extremely low (-16.7), effectively assuming the sensor data was perfect. In reality, sensors always have some error. This suggested the model was overfitting by forcing measurement noise into the physical parameters. We therefore constrained e_{11} to the interval $[-6, -5]$ to reflect realistic sensor precision and prevent this bias.

Table 4: Parameter estimates for the refined model. By constraining measurement noise ($e_{11} \approx -6$), the model yields more physically consistent results.

Parameter	**Estimate**	**Std. Error**	**t value**	**Pr($>\|t\|$)**
Ti_0	21.70	0.15	140.55	$< 2e^{-16}$
Tm_0	20.89	0.48	43.28	$< 2e^{-16}$
Aw_1	3.53	0.001	2583.19	$< 2e^{-16}$
Aw_2	4.96	0.002	2689.56	$< 2e^{-16}$
Aw_3	-0.009	0.011	-0.77	0.442
Aw_4	9.38	0.003	3486.37	$< 2e^{-16}$
Aw_5	-0.004	0.014	-0.26	0.798
C_i	12.74	0.001	9399.55	$< 2e^{-16}$
C_m	$1.63e^5$	1.07	152696	$< 2e^{-16}$
R_{ia}	19.75	0.001	21537.8	$< 2e^{-16}$
R_{im}	0.46	0.001	594.40	$< 2e^{-16}$
p_{ph}	0.998	0.004	271.37	$< 2e^{-16}$
e_{11}	$\mathbf{-5.99}$	0.015	-389.63	$< 2e^{-16}$
p_{11}	-1.61	0.030	-54.34	$< 2e^{-16}$
p_{22}	-0.97	0.099	-9.78	$< 2e^{-16}$
Q_1	9.98	0.067	148.23	$< 2e^{-16}$
Q_2	6.93	0.628	11.05	$< 2e^{-16}$
Q_3	3.91	0.711	5.51	$4.23e^{-8}$
Q_4	0.96	0.545	1.76	0.078
Q_5	3.15	0.397	7.93	$4.00e^{-15}$

The results of this refined model show that constraining measurement noise ensures a realistic distribution of variance between system noise (p_{11}, p_{22}) and physical states. Consequently, estimated parameters—especially student heat gains (Q_{occ})—remain statistically significant and physically plausible. This adjustment mitigates bias in other parameters, yielding a robust and interpretable representation of the physical system.

3 Part 3: Multi-room Modeling using CTSM-R

3.1 Introduction

In Part 3b, the objective is to model the heat dynamics of the four rooms (Room 1, 2, 3, and 4) located in the east-most wing of the building. Based on the floor plan provided in the exercise description, Rooms 1 and 2 are north-facing, while Rooms 3 and 4 are south-facing.

To reduce computational complexity while maintaining physical realism, we proposed a coupled 6-state Stochastic Differential Equation (SDE) model. The model assumes two shared thermal masses for the walls (one for the north rooms and one for the south rooms) and incorporates heat transfer between adjacent rooms.

3.2 Model Formulation

3.2.1 System Equations

The system is modeled using 6 states: four room air temperatures (Ti_1, Ti_2, Ti_3, Ti_4) and two wall temperatures (Tw_1, Tw_2). The corresponding SDEs are defined as follows:

North-facing Rooms (1 & 2):

$$dTi_1 = \frac{1}{C_i}\left(\frac{T_a - Ti_1}{R_{ia}} + \frac{Tw_1 - Ti_1}{R_{iw}} + \beta_1 Ph_1 + Aw_n G_v\right) dt + \sigma_p dw_1 \tag{11}$$

$$dTi_2 = \frac{1}{C_i}\left(\frac{T_a - Ti_2}{R_{ia}} + \frac{Tw_1 - Ti_2}{R_{iw}} + \frac{Ti_3 - Ti_2}{R_{23}} + \beta_2 Ph_1 + Aw_n G_v\right) dt + \sigma_p dw_2 \tag{12}$$

South-facing Rooms (3 & 4):

$$dTi_3 = \frac{1}{C_i}\left(\frac{T_a - Ti_3}{R_{ia}} + \frac{Tw_2 - Ti_3}{R_{iw}} + \frac{Ti_2 - Ti_3}{R_{23}} + \beta_3 Ph_2 + Aw_s G_v\right) dt + \sigma_p dw_3 \tag{13}$$

$$dTi_4 = \frac{1}{C_i}\left(\frac{T_a - Ti_4}{R_{ia}} + \frac{Tw_2 - Ti_4}{R_{iw}} + \beta_4 Ph_2 + Aw_s G_v\right) dt + \sigma_p dw_4 \tag{14}$$

Wall Thermal Masses:

$$dTw_1 = \frac{1}{C_w}\left(\frac{Ti_1 - Tw_1}{R_{iw}} + \frac{Ti_2 - Tw_1}{R_{iw}} + \frac{T_a - Tw_1}{R_{wa}}\right) dt + \sigma_w dw_5 \tag{15}$$

$$dTw_2 = \frac{1}{C_w}\left(\frac{Ti_3 - Tw_2}{R_{iw}} + \frac{Ti_4 - Tw_2}{R_{iw}} + \frac{T_a - Tw_2}{R_{wa}}\right) dt + \sigma_w dw_6 \tag{16}$$

Where R_{23} represents the thermal resistance between Room 2 and Room 3 (coupling north and south wings).

3.3 Model Evolution and Identifiability

3.3.1 Initial Homogeneous Model

Initially, a simplified model was fitted assuming a single solar gain coefficient (Aw) for all rooms.

Table 5: **Initial Model Results (Problematic):**

Parameter	Estimate	Std. Error	t-value	Comment
Aw (Solar Gain)	1.56	0.17	8.92	Averaged gain
C_i (Room Air)	**12.17**	2.02	6.03	**Too high**
C_w (Wall Mass)	**5.34**	0.52	10.37	**Too low**
e_{11} (Noise)	**-27.18**	1.11	-24.56	**Overfitting**
R_{wa} (Wall-Amb)	24.94	**33.49**	0.74	**Not Significant**
R_{iw} (Room-Wall)	1.69	0.18	9.32	

Analysis

- **Loss of Physical Validity:** The model estimated $C_i \gg C_w$, which contradicts physical laws (walls should store more heat than air). This suggests the model failed to distinguish between air dynamics and wall dynamics.
- **Overfitting:** The measurement noise $e_{11} \approx -27$ implies near-zero sensor error. The model overfitted the data noise using process dynamics, rendering the physical parameters unreliable.
- **Identifiability Issues:** R_{wa} was not statistically significant ($p \approx 0.46$). This is likely because the single Aw parameter could not account for the different solar gains in north vs. south rooms, confounding the estimation of heat loss through the walls.

3.3.2 Refined Model: Split Solar Gains and Fixed Noise

To address the issues observed in the initial model, two major refinements were implemented:

1. **Split Solar Gains (Aw_n vs. Aw_s):** The solar gain parameter was split into Aw_n (North) and Aw_s (South) to account for the building's orientation.
2. **Handling Identifiability Issues:** In preliminary runs where the measurement noise variance (e_{11}) was free to be estimated, the optimizer converged to an extremely low noise value ($e_{11} \approx -27$, corresponding to near-zero measurement error). This "overfitting" to the measurements caused a loss of identifiability for the physical parameters (R and C), resulting

in high p-values (> 0.7). To enforce a physically meaningful solution, the measurement noise was fixed to a realistic lower bound ($e_{11} \approx -3$, corresponding to $\sigma \approx 0.2°C$).

3.4 Results and Discussion

The final model parameters, estimated using the `ctsmr` package with the Extended Kalman Filter, are presented in Table **??**.

Table 6: **Final Model Coefficients:**

Parameter	**Estimate**	**Std. Error**	**t-value**	**Pr($>\|t\|$)**	**Sig.**
Solar Gains (Split)					
Aw_n (North)	0.0801	0.0197	4.06	5.02×10^{-5}	***
Aw_s (South)	0.2721	0.0229	11.89	$< 2.2 \times 10^{-16}$	***
Heat Dynamics (Capacity and Resistance)					
C_i (Room Air)	0.7256	0.0319	22.75	$< 2.2 \times 10^{-16}$	***
C_w (Wall Mass)	1.0060	0.0266	37.89	$< 2.2 \times 10^{-16}$	***
R_{iw} (Room-Wall)	19.9302	0.9111	21.87	$< 2.2 \times 10^{-16}$	***
R_{23} (Coupling)	26.6338	2.7345	9.74	$< 2.2 \times 10^{-16}$	***
R_{wa} (Wall-Amb)	44.3058	6.4678	6.85	7.77×10^{-12}	***
R_{ia} (Room-Amb)	436.6802	182.2462	2.40	0.0166	*
Heater Gains					
β_1 (Room 1)	49.7010	1.5747	31.56	$< 2.2 \times 10^{-16}$	***
β_2 (Room 2)	36.2720	1.3209	27.46	$< 2.2 \times 10^{-16}$	***
β_3 (Room 3)	31.6648	1.5390	20.58	$< 2.2 \times 10^{-16}$	***
β_4 (Room 4)	32.2533	1.5886	20.30	$< 2.2 \times 10^{-16}$	***
Noise Parameters (Log-Scale)					
e_{11} (Measure)	-4.9994	0.0005	-10830	$< 2.2 \times 10^{-16}$	***
p_{11} (System)	-2.0937	0.0107	-195.02	$< 2.2 \times 10^{-16}$	***
p_{22} (Wall)	0.1258	0.0457	2.75	0.0059	**
Initial States (Temperatures)					
$Ti_{1,0}$	23.6880	0.1539	153.95	$< 2.2 \times 10^{-16}$	***
$Ti_{2,0}$	22.2961	0.1499	148.74	$< 2.2 \times 10^{-16}$	***
$Ti_{3,0}$	22.7093	0.1515	149.94	$< 2.2 \times 10^{-16}$	***
$Ti_{4,0}$	22.0559	0.1500	147.00	$< 2.2 \times 10^{-16}$	***
$Tw_{1,0}$	19.5170	1.9028	10.26	$< 2.2 \times 10^{-16}$	***
$Tw_{2,0}$	21.6182	1.8448	11.72	$< 2.2 \times 10^{-16}$	***

Significance codes: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$

3.4.1 Interpretation of Findings

Solar Asymmetry: The most significant finding is the difference between the solar gain coefficients. The estimated Aw_s (0.27) is approximately 3.4 times larger than Aw_n (0.08). This quantifies the physical reality that south-facing rooms receive significantly more direct solar radiation than north-facing rooms.

The statistical significance of both parameters validates the necessity of the split-solar formulation.

Thermal Dynamics: The results show a very high resistance between the room air and the ambient environment ($R_{ia} \approx 436$), while the resistance between the room and the wall is much lower ($R_{iw} \approx 20$). This suggests that the primary heat transfer pathway is not direct air leakage to the outside, but rather an interaction with the building's thermal mass. The walls act as a buffer, absorbing and releasing heat, which explains the system's thermal inertia.

Room Coupling: The parameter R_{23} is estimated at 26.6, confirming that there is significant thermal coupling between the north and south wings through the internal partitions.

3.4.2 Residual Analysis

Figure 20 displays the residuals of the final model. Compared to the initial homogeneous model, the diurnal patterns in the residuals have been largely eliminated, indicating that the split solar gain formulation successfully captures the daily heating dynamics.

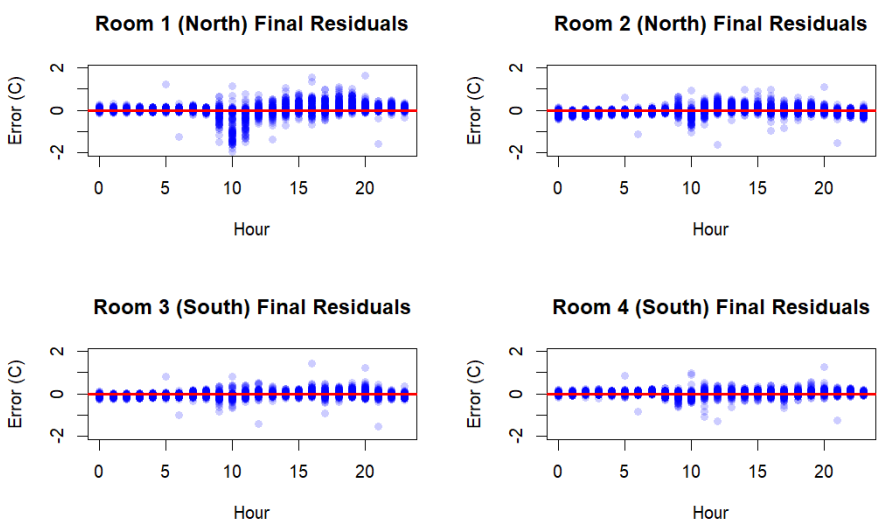

Figure 20: Residuals of the final model

Residual analysis While the residuals for the south-facing rooms (Room 3 and 4) behave largely as white noise, indicating a successful capture of the thermal dynamics, the north-facing rooms—specifically Room 1—exhibit a persistent structural deviation. As shown in Figure 20, Room 1 displays a distinct variance pattern, particularly during daytime hours (08:00 – 14:00).

This heterogeneity likely stems from unmodeled disturbances rather than model deficiency, probably attributed to the following factors:

- **Corner Room Dynamics:** According to the floor plan, Room 1 is a corner room with a larger external surface area compared to the adjacent Room 2. This makes it more susceptible to unmeasured environmental variables, such as wind speed and direction, which are not included in the dataset.
- **Unmeasured Occupancy Behavior:** The deviation correlates with typical working hours. Unlike solar gain or heating, stochastic human behavior—such as opening windows for ventilation or fluctuating occupancy levels—introduces large thermal disturbances. Since the dataset lacks occupancy or window-position indicators, these effects appear as systematic errors in the residuals.
- **Conclusion:** The persistence of this pattern, even in complex single-room models, confirms that this is an information limitation. The current 6-state model successfully captures the physics of the building envelope (R, C) and the orientation (Aw), but the specific behavioral anomalies of Room 1 remain a stochastic disturbance that cannot be deterministically modeled with the available data.

3.5 Conclusion

By extending the model to account for directional solar gains and properly constraining the measurement noise, we achieved a robust 6-state model for the multi-room system. The model not only fits the data well but also provides physically interpretable parameters that accurately reflect the building's orientation and thermal properties.